

Qatar-1b

R-2787

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140623 · created 2026-09-04T00:38:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456831.93367 +/- 0.00032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.053
Transit depth (Rp/Rs)^2	2.77 +/- 1.77 [%]
Orbital Inclination (inc)	89.03 +/- 2.81
Airmass coefficient 1 (a1)	11.54 +/- 1.08
Airmass coefficient 2 (a2)	-2.05 +/- 0.71
Scatter in the residuals of the lightcurve fit is	9.25 %
Best Comparison Star	#3 - [483, 236]
Optimal Aperture	1.88
Optimal Annulus	10.01
Transit Duration (day)	0.0784 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.053 vs 0.1463 (deviation 0.37σ)
Duration fit vs published	0.0784 d vs 0.0692 d (deviation 1.64σ)
Mid-time O–C	0.4 min
Depth SNR	3.1
Residual scatter	9.25 %

Light curve

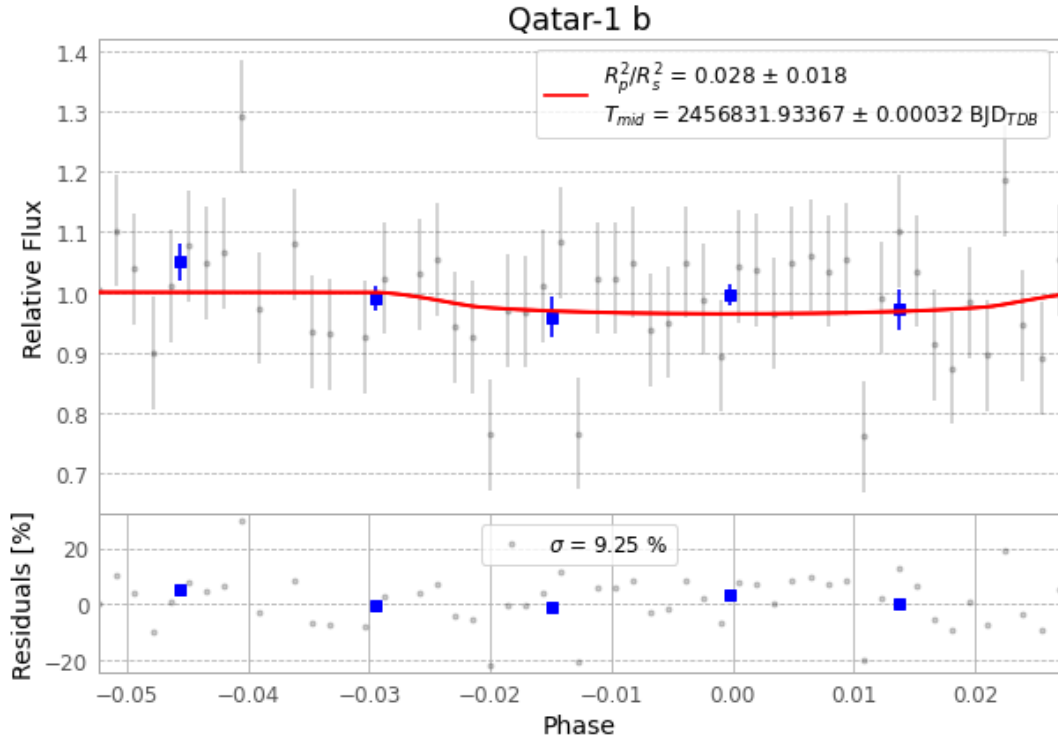


Figure 1 — FinalLightCurve_Qatar-1 b_2014-06-23.png (SHA-256 6ed089b478c1f79e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 292], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c1aabe073bd04ab3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), Qatar-1140623083613.FITS → Qatar-1140623111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140623.log (e9bbf46b9143...), FinalLightCurve_Qatar-1 b_2014-06-23.png (6ed089b478c1...), FinalLightCurve_Qatar-1 b_2014-06-23.csv (5b00978ce031...)

Compute resources

Wall time 156.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:38Z.